

Vandan Agrawal

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Software Engineer with production experience across full-stack development, RAG/LLM systems, and security principles. Builds and ships end-to-end, with a track record of measurable impact across startups and security research.

Education

University of Florida

December 2025

Master of Science in Computer Science | GPA: 3.66

Amity University

May 2024

Bachelor of Technology in Computer Science and Engineering

Coursework: Distributed OS Principles, Advanced Computer Networks, Programming Language Principles, Trustworthy Machine Learning, Malware Reverse Engineering, Intro to Cryptology, Cyber-Physical Systems Security

Skills

- **Languages:** Python, TypeScript, Java, C++, C, Kotlin, SQL, x64 assembly
- **Frameworks and Libraries:** Next.js, Flutter, Tkinter, Firebase, Groq Cloud, FastAPI
- **Developer Tools:** Google Cloud Platform, Claude, Git, AWS, Vertex AI
- **Security tools:** Ghidra, x64dbg, VMware, Wireshark, Splunk
- **Operating Systems:** Windows, Linux (Debian, Kali, Ubuntu), MacOS

Experience

Software Engineer, Skedaddle Enterprises

March 2026 – Present

- Architected a low-latency RAG-based AI system serving 5,000+ users using Llama 3.3 and Groq API; queried Firestore vector embeddings to generate real-time personalized Context-Cards and event recommendations.
- Migrated 8 Swift and Flutter screens to Kotlin, unifying the Android stack and cutting deployment overhead.
- Engineered location-aware event discovery pipeline by integrating Ticketmaster and Google Places APIs.
- Built real-time RSVP/notification system with NoSQL Firestore listeners, supporting 10000+ concurrent users.
- Aligned API contracts across Android and Core infrastructure teams to maintain cross-platform logic parity.

Cybersecurity Analyst, Malware Reverse Engineering Lab

January 2025 – May 2025

- Reverse-engineered over 10 real-world malware samples analyzing direct system calls in x64 assembly to identify execution flow, persistence, C2, and IOCs, enabling rapid incident response against threats.
- Ran 60+ hours of static/dynamic analysis using Ghidra, Wireshark, and x64dbg in a multi-system sandbox.
- Reduced manual reverse-engineering time 90% by automating obfuscation-bypass with Python scripts.
- Dissected a variant of the Akira ransomware as a final project, documenting behavior and IOCs for a threat that has impacted over 250 small to medium-sized American organizations.
- Authored YARA rules for malware family detection, strengthening endpoint security posture.

Software Developer Intern, SmartInternz

May 2023 – July 2023

- Deployed 20+ Salesforce objects/workflows enforcing secure, role-based access control.
- Built async Apex triggers/batch processes for high-concurrency data, improving responsiveness 15%.
- Automated compliance monitoring for 5000+ records via Shield Encryption, cutting manual review by 60%.
- Orchestrated 500+ weekly transactions via REST API integrations and Screen Flows.

Projects

BinSight - .exe Inspection Web App - React, TypeScript, Fastify, Docker

September 2026

- Built a full-stack malware analysis dashboard parsing files entirely client-side in a sandboxed Web Worker.
- Shipped a 6-gate CI/CD pipeline using GitHub Actions, catching 2 critical CVEs before release.
- Hardened on Oracle Cloud with dual-mode AI reporting, maintaining 99.9% uptime across live audits.
- Audited the live deployment, patching a spoofable rate limiter, missing CSP headers, and an XSS sanitizer gap.

Clinical Trial Immune Cell Analytics Dashboard - Python, SQLite, FastAPI

August 2026

- Designed normalized SQL schema for clinical trial dataset consisting of 3,500 subjects and 52,500 records.
- Enforced data integrity via composite unique constraints, preventing cross-project subject ID collisions.
- Authored production scaling plan for SQLite to Postgres migration and query precomputation as trial data grows.

Municipal Research & Analysis Engine - Google Cloud Platform (GCP), Claude Code, TypeScript, Python

June 2026

- Architected multilingual LLM search engine on GCP with Next.js/FastAPI, adding OAuth 2.0 and rate-limiting.
- Developed RAG pipeline with Vertex AI Search and Gemini 2.5, automating daily incremental ingestion from New York City legislative APIs to keep 8,000+ municipal documents synchronized and searchable
- Added Google OAuth and Firestore-backed notebook for persistent bookmarking and search history.

Distributed Reddit Concurrency Engine and Simulator - Go (Golang)

December 2024

- Designed a lock-free, actor-based distributed system in Go (Proto.Actor) simulating a Reddit backend with concurrent user registration, subreddit management, post voting, karma calculation, and direct messaging.
- Modeled user activity via Zipf distribution to simulate non-uniform subreddit membership and posting.
- Benchmarked performance up to 100,000 concurrent actors, profiling memory, and goroutines using Go pprof.